A Bayesian Approach to Modeling Infection-Based Social Distancing in the SARS-CoV-2 Pandemic

Ulyana Piterbarg[a]

[a]*Mathematics Department & Institute for Data, Systems, and Society (IDSS), MIT*

1. Introduction

Compartmental models are a popular class of models used in mathematical epidemiology for the study of disease transmission. Models belonging to this class simplify the complexity of real-world disease transmission dynamics by assuming that a population is well-mixed, similarly to mean-field models in physics and chemistry which approximate high dimensional interactions via averaging under the principle of mass action.

The simplest compartmental model is the *Susceptible-Infective-Removed* (*SIR*) model, which parameterizes disease transmission dynamics between susceptible, infected, and recovered (or dead) individuals in a population as a system of non-linear ordinary differential equations (ODEs) governed by the fixed constants β, λ, and $\mathcal{R}_0$, corresponding respectively to the *mass action* parameter, the *infective removal rate*, and the *basic reproduction number* [4, 14]. One major failure of the classical *SIR* model, however, is that its representation of β, λ, and $\mathcal{R}_0$ as constants makes it agnostic to the potential effects of societal response to known infection and deaths on dynamics. In response, researchers have augmented the SIR model with additional terms designed to capture behavioral effects witnessed in previous pandemics [5, 9, 10].

Throughout the SARS-CoV-2 pandemic, governments across the world have used infection information to regulate social distancing behaviors as well as viral transmission rates, allowing policymakers to "control the course of the pandemic as it unfolds." The result is a *negative feedback* effect, with average daily interactions between susceptible and infected populations proportionate to known infection rates. Franco has proposed capturing the effects of infection-based social distancing policy on transmission dynamics via the addition of a negative-feedback term to the *SIR* model, yielding a new model for disease transmission tailored to the SARS-CoV-2 pandemic, feedback-*SIR* (f*SIR*). In contrast to the classical *SIR* model, f*SIR* represents the average transmission rate as a function of time, rather than as a fixed constant [6].

In this paper, we use a Bayesian approach to model and analyze the effects of various governments' infection-based social distancing policies during the SARS-CoV-2 pandemic by fitting the f*SIR* model with *linear distancing* to observed disease transmission statistics via the PyMC3 probabilistic programming framework. First, we present the assumptions and characteristics of the *SIR* and f*SIR* models in more detail. Then, we discuss our experimental methodology for using Bayesian inference to fit the f*SIR* model with linear distancing to SARS-CoV-2 pandemic data. We also analyze the fit of the model across four different countries with SARS-CoV-2 pandemics (China, France, the United States, and Japan), as well as our resultant, inferred posterior distributions for model parameters. Lastly, we provide a comparative analysis of various governments' policy responses during the early phases of pandemics and the corresponding, estimated SARS-CoV-2 transmission rates over time derived from the fitted f*SIR* models.

2. *SIR* and f*SIR*

2.1. Definitions

As previously mentioned, *compartmental models* of disease transmission assume that a population is well-mixed and can be divided across non-overlapping groups, or compartments. We then make a series of assumptions about the rates of transfer between compartments. Indeed, the independent variable in compartmental models is time, t, and the rates of transfer between compartments are expressed as time derivatives of the compartments' sizes. In f*SIR*, as in classical *SIR*, the population is divided into exactly three classes S, I, R.

1. S - the susceptible class. The group of people who are susceptible to infection, never having been exposed to the disease.
2. I - the infective class. The group of people who can infect others with the disease, whether symptomatic or asymptomatic.

3. R - the removed class. Those who have successfully recovered from the disease (and are now immune) as well as removed dead bodies.

We model the number of people in each class as a function of t, denoted as $S(t), I(t), R(t)$, respectively.

Next, we define the following set of epidemiological terms, which we split into two groups: (1) those that are relevant to both the *SIR* and f*SIR* models and (2) those that are relevant only to f*SIR*.

(1) Both *SIR* and f*SIR*

- *mass action incidence*, βN: the number of people an average member of the population makes contact with per unit time; here, we only consider contact that is sufficient to transmit the disease.
- *infective removal rate*, λI: the rate at which people leave the infective class I per unit time.
- *basic reproduction number*, $\mathcal{R}_0 = \frac{\beta S(0)}{\lambda}$: the number of susceptible individuals in class S that an average member of the infective class I infects per unit time.

(2) Only f*SIR*

- *distancing function*, $\kappa(I)$: average population response to knowledge of current infection numbers I in relation to typical interaction patterns, satisfying $\kappa(I) \geq 0, \forall I$. Large, positive values of $\kappa(I)$ indicate a highly reactive social-distancing response to current infection numbers, while values of $\kappa(I)$ closer to 0 indicate a less reactive social-distancing response.
- *effective reproduction number*, $\mathcal{R}(t)$: the number of susceptible individuals in class S that an average member of the infective class I infects per unit time when infection-aware social distancing policies are accounted for. The effective reproduction number varies as a function of time, is indirectly related to the distancing function $\kappa(I)$, and is directly related to the basic reproduction number $\mathcal{R}_0 = \frac{\beta S(0)}{\lambda}$, with $\mathcal{R}(0) = \mathcal{R}_0$.

2.2. Assumptions

In modeling disease transmission dynamics, the *SIR* and f*SIR* models rely upon following central assumptions concerning the total population:

1. All individuals in the population belong to exactly one compartment S, I, or R.
2. All individuals are equally likely to make contact.
3. There is no entry into, or departure from the population during the pandemic, except possibly through death due to the disease. In other words, the total population N is constant over time (i.e. $\frac{d}{dt}N = 0$).
4. At the start of an epidemic, all individuals in the population are susceptible, $S(0) = N$.

2.3. *SIR*

If we consider the previously defined terms in group (1) in light of the assumptions described above, we can formulate a system of non-linear ODEs representing the dynamics between the compartments S, I, R and corresponding to the governing equations of the *SIR* model.

First, note that under the first and third assumptions, it must be true that $N = S(t) + I(t) + R(t), \forall t \geq 0$.

Second, observe that the number of individuals in the susceptible class will only decrease over time, assuming that $\exists t \geq 0 : I(t) > 0$ (i.e. there is at least one infected individual at some time t). Moreover, with the *mass action incidence* given by βN and the probability that an infected individual makes contact with a susceptible individual given by $\frac{S}{N}$ for all times t (under assumption 2), we can calculate the number of new infections per infective, per unit time as $\beta N \cdot \frac{S}{N} = \beta S$. Thus, the rate of new infections per unit time must be βSI.

Third, let's consider the change in the total number of individuals in the infective class per unit time. As derived above, new people are transferred from S to I with positive rate βSI. However, under our definition of the *infective removal rate*, we know that the rate at which people leave the class I for the class R per unit time is given by λI. Thus, the total rate of change in I per unit time is given by $\beta SI - \lambda I$.

Finally, under assumptions 1 and 3, we know that the change per unit time in the number of individuals in the removed class depends only on the rate of transfer from I to R, given by λI.

$$\begin{aligned} \dot{S} &= -\beta SI \\ \dot{I} &= \beta SI - \lambda I \\ \dot{R} &= \lambda I \\ N &= S(t) + I(t) + R(t), \forall t \geq 0 \end{aligned} \tag{1}$$

Note that under the last equation above, the *SIR* model can be reduced to a system of just two non-linear ODEs dependent on susceptibles over time and infectives over time. Moreover, we can normalize each of our variables by setting $s(t) = S(t)/N, i(t) = I(t)/N$, and $r(t) = R(t)/N$. We can also replace the parameter β with $\mathcal{R}_0\lambda$, given that we have $\mathcal{R}_0\lambda = \beta s(0) = \beta$.

$$\begin{aligned} \dot{s} &= -\mathcal{R}_0\lambda si \\ \dot{i} &= \mathcal{R}_0\lambda si - \lambda i \\ 1 &= s(t) + i(t) + r(t), \forall t \geq 0 \end{aligned} \tag{2}$$

The simplified, normalized form of the *SIR* model's governing equations given in Eq (2) is used by epidemiologists more often in practice than the full set of unnormalized equations given in Eq (1), particularly when it comes to performing comparative studies of disease transmission dynamics, which is the focus of this paper [4, 10, 14].

2.4. f*SIR*

As described in the introduction, the principal difference between the *SIR* model and f*SIR* model proposed by Franco is the fact that the latter includes an additional term that accounts for the infection-based social distancing policies enacted by countries around the globe during the SARS-CoV-2 pandemic.

Practically, the effects of infection-based social distancing policies are re-parameterized in the f*SIR* model by replacing the basic reproduction number $\mathcal{R}_0$ in the governing equations of the *SIR* model with the dynamic and more sophisticated *effective reproduction number* $\mathcal{R}(t)$, which varies over time and is related to both the distancing function $\kappa(I)$ and the basic reproduction number $\mathcal{R}_0$. Since the number of infective individuals is a function of time and we are interested in determining the f*SIR* governing equation in *normalized* form, we rewrite our distancing function in normalized form with respect to time by composing it with the population proportion of infective individuals $i(t)$, yielding $(\kappa \circ i)(t)$.

$$\mathcal{R}(t) = \frac{\mathcal{R}_0}{1 + (\kappa \circ i)(t)} \tag{3}$$

The effective reproduction number introduces a *negative* feedback loop into the basic disease transmission dynamics model predicted by the *SIR*, capturing (1) "the fact that society decreases interactions in response to an increase of infections, thereby reducing the reproduction number," and (2) "the fact that society is expected to return to typical interaction patterns when infections are not present" [6].

We write the infection-aware governing equations of the f*SIR* model in normalized, reduced form below.

$$\begin{aligned} \dot{s} &= -\frac{\mathcal{R}_0\lambda si}{1 + (\kappa \circ i)(t)} \\ \dot{i} &= \frac{\mathcal{R}_0\lambda si}{1 + (\kappa \circ i)(t)} - \lambda i \\ 1 &= s(t) + i(t) + r(t), \forall t \geq 0 \end{aligned} \tag{4}$$

3. Experimental Methodology

3.1. Motivations

"Real-world" data reflecting disease transmission is inevitably noisy. In the SARS-CoV-2 pandemic, sources of noise effecting data of known infections, recoveries, and deaths are plentiful. A list of several such sources of noise (by no means exhaustive) is provided below.

- *Undertesting.* Particularly in the early stages of the epidemic, testing supplies were scarce in many countries, making it practically impossible to test all individuals in a population to identify those infected with the coronavirus. Even as the pandemic has progressed and production of tests has increased around the world, widespread testing is not a reality in many countries with larger populations.
- *Asymptomatic, or mildly symptomatic cases.* It is known that there are infected individuals who are carriers for SARS-CoV-2 despite displaying few, if any symptoms. In present day, the occurrence percentage of such cases is still disputed, though some initial studies suggest their extent may be formidable [2].

- *Under reporting of true data.* We may suspect that some countries may under report true statistics of disease transmission for various objectives.

One of the most effective and widely used approaches in the epidemiological literature for fitting models to noisy disease transmission data is Bayesian inference, and particularly Markov Chain Monte Carlo (MCMC) methods [1, 8, 12].

Thus, we choose to infer posterior distributions for parameters of the f*SIR* model directly from data via Hamiltonian Monte Carlo (HMC) [11], as implemented in the probabilistic programming framework PyMC3 [13]. A description of the HMC algorithm is provided in appendix A.

3.2. Accounting for Noise

Before we begin inference, it is important to explicitly account for the aforementioned noise in our data, whether resultant from the data collection process itself or from outside factors disrupting the idealized assumptions of the f*SIR* model.

To that end, we make a distinction between $s_{\text{obs}}(t), i_{\text{obs}}(t), r_{\text{obs}}(t)$, corresponding respectively to the *reported proportions* of susceptible, infective, and removed individuals in a population, and the *true underlying functions* $s(t), i(t), r(t)$, whose dynamics are described by the normalized and reduced governing equations of the f*SIR* model (Eq (4)).

Indeed, we assume that for each fixed time, t, we have:

$$\begin{aligned} s_{\text{obs}}(t) &\sim \mathcal{N}(s(t), \widetilde{\sigma}^2) \\ i_{\text{obs}}(t) &\sim \mathcal{N}(i(t), \widetilde{\sigma}^2) \\ r_{obs}(t) &= 1 - s_{obs}(t) - i_{obs}(t), \end{aligned} \tag{5}$$

where $\widetilde{\sigma}$ is some unknown random variable.

We also account for noise by choosing the simplest form possible for our *distancing function* $(\kappa \circ i)(t)$ in f*SIR*:

$$(\kappa \circ i)(t) = \kappa \cdot i(t) \implies \mathcal{R}(t) = \frac{\mathcal{R}_0}{1 + \kappa \cdot i(t)} \tag{6}$$

Thus, we model infection-aware social distancing policies as *directly proportional* to the normalized number of infective individuals i in each population we consider, parameterized by an unknown *distancing constant* κ.

Along with β, λ, and $\mathcal{R}_0$, we will construct estimated posterior distributions for our noise parameter $\widetilde{\sigma}$ as well as our distancing constant κ via HMC-based Bayesian inference.

3.3. Choosing Priors for Inference

In order to make our Bayesian inference problem tractable, we estimate the distributions of our f*SIR* parameters of interest according to the following continuous priors:

$$\begin{aligned} \beta &\sim \text{Unif}[0, 0.6] \\ \lambda &\sim \text{Unif}[1/18, 1/9] \\ \kappa &\sim \text{Unif}[0, 10^4] \\ \widetilde{\sigma}^2 &\sim \text{HalfCauchy}(\mu = 0, \gamma = 0.25) \end{aligned}$$

Note that we do not initialize a prior for the basic reproduction prior since by definition, $\mathcal{R}_0 = \beta/\lambda$ in the f*SIR* model (just as in the *SIR* model). Consequently, the estimated posterior of $\mathcal{R}_0$ is defined deterministically as the quotient of the estimated posteriors of β and λ.

In selecting the bounds of our uniform priors for β, λ, and κ, we follow the conventions established by Franco in their regression experiments with the f*SIR* model [6].

We choose the Half-Cauchy distribution centered on zero with $\gamma = 0.25$ as our prior for the noise parameter $\widetilde{\sigma}^2$ due to the fact that this distribution has most of its mass concentrated around 0 with little standard deviation. This reflects our prior belief that variance of $s_{\text{obs}}(t), i_{\text{obs}}(t)$ away from idealized, normalized disease transmission dynamics $s(t), i(t)$ predicted by f*SIR* must be positive and will have small magnitude expectation, $0 \leq \mathbb{E}[\widetilde{\sigma}^2] << 1$.

3.4. Data Processing

We choose the following four countries for our analysis: (1) China, (2) France, (3) United States, and (4) Japan.

We obtain time series data representing daily totals for reported numbers of confirmed cases, recovered patients, and deaths for the SARS-CoV-2 epidemics in our four countries of interest as of May 11th, 2020 via the *2019 Novel Coronavirus SARS-CoV-2 (2019-nCoV) Data Repository*, managed by the Johns Hopkins University Center for Systems Science and Engineering, and made public for educational and academic research purposes.

As estimates of the total populations of our countries of interest, we employ the 2020 mean population estimates for each of the eight nations computed and released for public use by the Department of Economic and Social Affairs of the United Nations.

Given the above, we process our data to obtain $s_{obs}(t), i_{obs}(t)$ as initially defined in Section 3.2 by recombining the reported total numbers of confirmed cases, recovered patients, and deaths at each day t in each country via the expressions below:

$$\begin{aligned} s_{obs}(t) &= 1 - \frac{(\text{confirmed})}{N} \\ i_{obs}(t) &= \frac{(\text{confirmed}) - (\text{recovered}) - (\text{deaths})}{N} \end{aligned} \tag{7}$$

We conclude our processing procedure by eliminating the long left-wise tails of our two sets of time series data, which correspond to the early periods of all four epidemics during which the transmission of SARS-CoV-2 was still fairly negligible, with fewer than 10^3 confirmed cases of the novel coronavirus yet detected. To that end, we discard all time-steps t of our time series $s_{obs}(t), i_{obs}(t)$ at which the total number of reported, confirmed cases fails to exceed 1000.

3.5. Training

We set up our inference model under the normalized, reduced governing equations of the f*SIR* model given in Eq (4) and the noise parameterization described in (3.2), instantiate our priors as described in (3.3), and process our data as described above in (3.4).

Next, we split our processed time series data $s_{obs}(t), o_{obs}(t)$ for each of our countries of interest into a training T_{train} and a testing set T_{test} according to the date $t^* = (\textit{May 4th, 2020})$.

$$\begin{aligned} T_{\text{train}} &= \{t : t \leq t^*\} \\ T_{\text{test}} &= \{t : t > t^*\} \end{aligned} \tag{8}$$

Since our full processed time series datasets include data up until May 11th, 2020, by splitting our training set from our testing set at t^*, we commit to withholding exactly one week of reported "susceptible" and "recovered" data from our model during training, to be used for model evaluation.

Next, we train four separate models on distinct training datasets T_{test}, each corresponding to a distinct country of interest, and obtain HMC traces for estimated posterior distributions of f*SIR* parameters $(\lambda, \beta, \mathcal{R}_0, \sigma^2, \kappa)$. Each model's HMC traces are computed with the PyMC3 `sample` function and are generated across 5000 random samples (or time-steps) run on two chains, with the first 1000 samples designated for "tuning" [13]. Path lengths are adaptively set in HMC with the No-U-Turn Sampler (NUTS), initialized via the `jitter+adapt_diag` method, and step-size is set to $\varepsilon = 0.25$ [7]. Refer to the attached code for more details.

3.6. Evaluating Model Fit

After training has concluded, we subject our models to a stricter convergence check than that implemented during training by HMC that involves our withheld testing set T_{test}.

Indeed, to determine adequate fit of the trained model to the original training data, we perform non-parametric bootstrap sampling on the HMC traces for our estimated parameter posterior distributions (with the "burn-in" period discarded) for $M = 3000$ total samples, obtaining M bootstrapped sample estimates for the values of the underlying f*SIR* curves $s(t), i(t)$ for all time-steps $t \in T = T_{test} \cup T_{train}$. Then, from our M bootstrapped posterior samples for each t, we compute the quantities $s^*_{0.005}(t), s^*_{0.995}(t), i^*_{0.005}(t), i^*_{0.995}(t)$, representing the bootstrapped 99.0% confidence intervals for the estimated, mean underlying values of the f*SIR* curves $s(t), i(t)$

at each time-step $t \in T$. Lastly, we evaluate adequate fit of our f*SIR* model via the test ψ based on our time series of 99.0% bootstrapped confidence intervals given below.

$$\begin{aligned} \psi &= \mathbb{1}(\psi_s \cdot \psi_i) \\ \psi_s &= \prod_{t \in T} \mathbb{1}\left(s_{obs}(t) \in [s^*_{0.005}(t), s^*_{0.995}(t)]\right) \\ \psi_i &= \prod_{t \in T} \mathbb{1}\left(i_{obs}(t) \in [i^*_{0.005}(t), i^*_{0.995}(t)]\right) \end{aligned} \tag{9}$$

In other words, $\psi = 1$ for some trained model if and only if both the training time series $\{s_{obs}(t), i_{obs}(t) | t \in T_{\text{train}}\}$ and the withheld time series $\{s_{obs}(t), i_{obs}(t) | t \in T_{\text{test}}\}$ *both* lie strictly inside the boostrapped 99% confidence intervals for the underlying f*SIR* curves $s(t)$ and $i(t)$ estimated from the trained model's inferred parameter posteriors.

We classify those countries for which the corresponding trained models pass the strict convergence test ψ to be *reasonably modeled* by the f*SIR* model of disease transmission with linear distancing, displaying *at least weak evidence of infection-based social distancing* and having *interpretable* posterior distributions for our model parameters. We only discuss *well modeled* countries in our comparative analysis of various governments' policy responses during the SARS-CoV-2 pandemic, interpreting and comparing the corresponding estimated parameter posterior distributions as well as the values of the boostrapped mean effective reproduction number $\mathcal{R}(t)$ for $t \in T = T_{\text{train}} \cup T_{\text{test}}$.

4. Results & Analysis

4.1. Model Fit and Parameter Estimates

4.1.1. China

In Fig.1, we include the observed transmission data $s_{obs}(t), i_{obs}(t)$ plotted against time series of boostrapped 99% confidence intervals for $s(t), i(t)$ estimated from our Bayesian inference model for the SARS-CoV-2 epidemic in China for all $t \in T$. These boostrapped confidence intervals are constructed as per the procedure described in section (3.6). We observe that the training data time series are indeed fully contained in both sets of 95% confidence intervals series, indicating that our model passes our criterion ψ for model fit, which we interpret as evidence for the fact that the model posterior estimates for the epidemiological parameters of the f*SIR* model *are interpretable*. Summary statistics for the estimated posterior distributions of model parameters, including boostrapped 95% confidence intervals for mean parameter values, are provided below 1.

Table 1: Estimated fSIR (with Linear Infection-Aware Distancing) Epidemiological Parameters for SARS-CoV-2 in China

Parameter	μ_{est}	95% Confidence Interval for μ_{est}
β	0.147	$[0.138, 0.150]$
λ	0.110	$[0.100, 0.111]$
$\mathcal{R}_0 = \beta/\lambda$	1.35	$[1.32, 1.39]$
κ	8670	$[5860, 9990]$
$\widetilde{\sigma}^2$	$2.45 \cdot 10^{-5}$	$[2.25 \cdot 10^{-5}, 2.73 \cdot 10^{-5}]$

Critically, observe that our fitted model estimates κ, our parameterized linear infection-based social distancing coefficient, to have a relatively high value $\hat{\kappa} \approx 8670$, indicating the predicted presence of a highly reactive social-distancing response. Moreover, the basic reproduction number $\mathcal{R}_0$ when there are no reported cases $t = 0, i(0) = 0$ is estimated as $\hat{\mathcal{R}}_0 \approx 1.35$, which we can interpret as indicating a fast lockdown response to the known presence of SARS-CoV-2 cases, with reported cases multiplying relatively slowly even at $t = 0$ as estimated from our processed SARS-CoV-2 transmission time series data. Finally, average recovery time is estimated as $(\hat{\lambda})^{-1} \approx 9.1$ days.

4.1.2. France

As in the case of our model for China, we can evaluate graphically that our Bayesian f*SIR* model with linear distancing passes our outlined model fit criterion ψ, with both training and withheld testing time series data contained in the boostrapped 99% confidence intervals computed from model posteriors, as shown in the convergence criterion ψ subplots for France included in Fig.1.

Table 2: Estimated fSIR (with Linear Infection-Aware Distancing) Epidemiological Parameters for SARS-CoV-2 in France

Parameter	μ_{est}	95% Confidence Interval for μ_{est}
β	0.275	$[0.252, 0.309]$
λ	$5.92 \cdot 10^{-2}$	$[5.90 \cdot 10^{-2}, 6.08 \cdot 10^{-2}]$
$\mathcal{R}_0 = \beta/\lambda$	4.67	$[4.19, 5.18]$
κ	2520	$[2040, 3150]$
$\widetilde{\sigma}^2$	$2.05 \cdot 10^{-4}$	$[1.76 \cdot 10^{-4}, 2.43 \cdot 10^{-4}]$

In comparing summary statistics for model estimates of epidemiological parameters for SARS-CoV-2 transmission and response in France and China (represented in Table 2 and Table 1 respectively), it is clear that our f*SIR*-based model estimates the reactivity of governmental policy response to reported SARS-CoV-2 cases in France to be *lower in magnitude* than the estimated reactivity of governmental policy response in China. Indeed, the average boostrapped value for the distancing constant κ corresponding to reported SARS-CoV-2 transmission data in France is only $\hat{\kappa} \approx 2520$, yielding a reactivity to new reported cases that is approximately a factor of three smaller in magnitude than the corresponding value estimated from SARS-CoV-2 data in China. Moreover, the basic reproduction number is estimated as $\hat{\mathcal{R}}_0 \approx 4.67$, more than two times higher in magnitude than the corresponding estimate based on reported transmission data in China. Similarly, average recovery time is estimated in France as $(\hat{\lambda})^{-1} \approx 16.9$ days, again nearly two times longer than average recovery time estimated in China.

4.1.3. US

Our model trained on SARS-CoV-2 disease transmission data for the United States – just like the corresponding models for China and France – passes the model fit test ψ, plots of which are shown in Fig.1.

Table 3: Estimated fSIR (with Linear Infection-Aware Distancing) Epidemiological Parameters for SARS-CoV-2 in the US

Parameter	μ_{est}	95% Confidence Interval for μ_{est}
β	0.315	$[0.289, 0.348]$
λ	$6.00 \cdot 10^{-2}$	$[5.92 \cdot 10^{-2}, 6.10 \cdot 10^{-2}]$
$\mathcal{R}_0 = \beta/\lambda$	5.31	$[4.91, 5.84]$
κ	1760	$[1370, 2180]$
$\widetilde{\sigma}^2$	$3.89 \cdot 10^{-4}$	$[3.32 \cdot 10^{-4}, 4.43 \cdot 10^{-4}]$

Observe that the US SARS-CoV-2 f*SIR* Bayesian inference model estimates the responsiveness of governmental policy to reported infections in the US as $\hat{\kappa} \approx 1760$, corresponding to a less dynamic response than that of either China or France. Furthermore, we have $\hat{\mathcal{R}}_0 \approx 5.31$, representing an initial reproduction number higher than inferred in either China or France as well, and $(\hat{\lambda})^{-1} \approx 16.7$ days, corresponding to an average recovery time much longer than that estimated in China, but nearly equal to that estimated in France.

4.1.4. Japan

Our final model, trained on reported SARS-CoV-2 disease transmission data for Japan also passes the basic model fit test ψ; thus, we conclude that our estimates for averaged, estimated epidemiological parameters are indeed interpretable.

Table 4: Estimated fSIR (with Linear Infection-Aware Distancing) Epidemiological Parameters for SARS-CoV-2 in the Japan

Parameter	μ_{est}	95% Confidence Interval for μ_{est}
β	0.148	$[0.136, 0.152]$
λ	$6.01 \cdot 10^{-2}$	$[5.87 \cdot 10^{-2}, 6.24 \cdot 10^{-2}]$
$\mathcal{R}_0 = \beta/\lambda$	2.44	$[2.26, 2.59]$
κ	7870	$[4930, 10000]$
$\widetilde{\sigma}^2$	$1.59 \cdot 10^{-5}$	$[1.42 \cdot 10^{-5}, 1.78 \cdot 10^{-5}]$

In Japan, our f*SIR* linear infection-aware distancing model estimates $\kappa, \mathcal{R}_0$ to be closer in value to those estimated based on reported data from China. Indeed, as shown in Table 4, we have $\hat{\kappa} \approx 7870$ and $\mathcal{R}_0 \approx 2.44$, representing a slightly less reactive infection-based response and a slightly higher initial reproduction number. With respect to recovery time, we compute $(\hat{\lambda})^{-1} \approx 16.4$ days – nearly equal to the corresponding estimates in the US and France, though significantly shorter than the estimate in China.

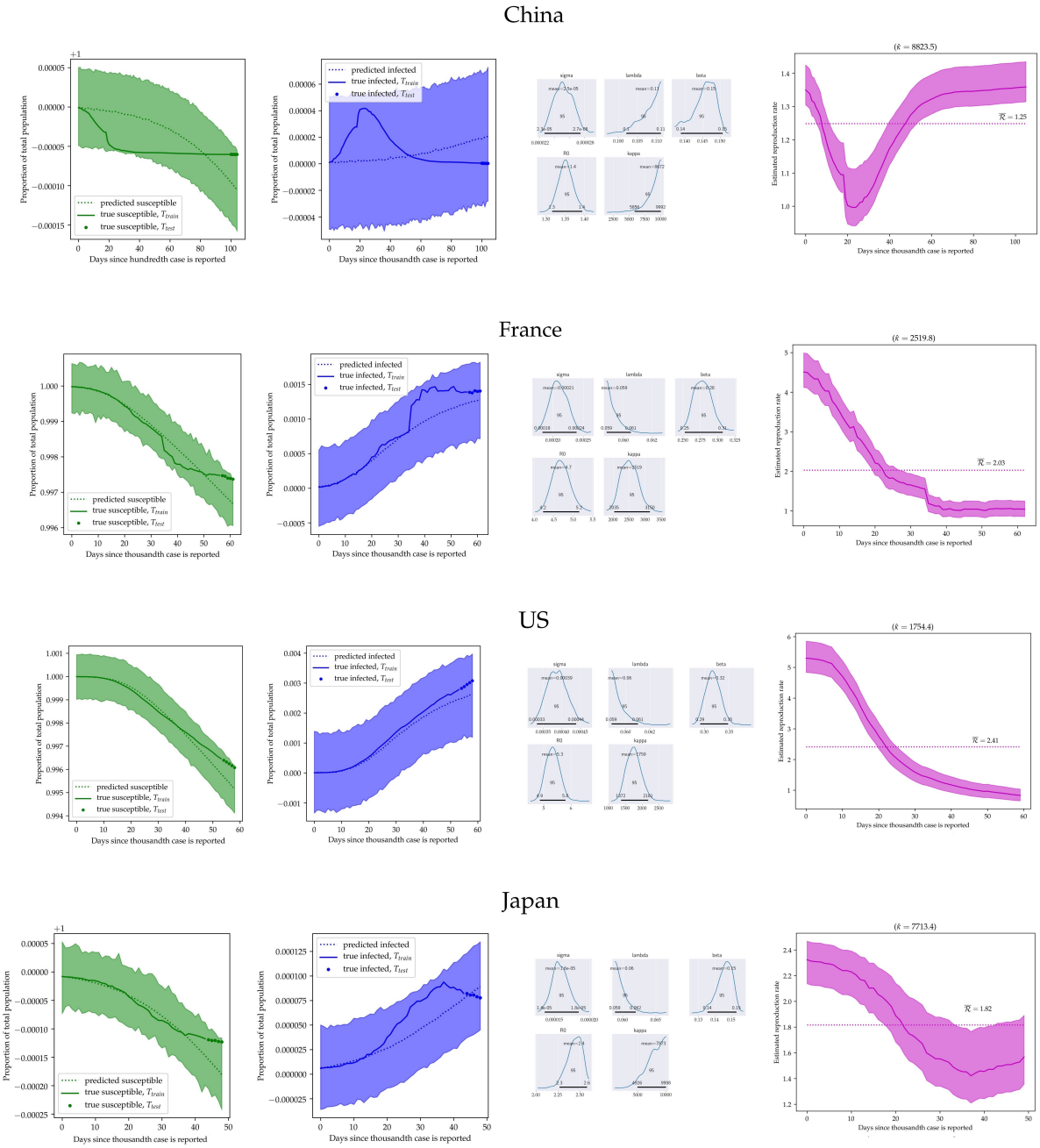

Figure 1: Evaluating inferred Bayesian model for the SARS-CoV-2 epidemic in China against the original training data and inspecting estimated posteriors as well as mean boostrapped values for effective reproduction number over time $\mathcal{R}(t)$. ***[Leftmost column] Inferred fSIR curves for $s(t), i(t)$ in the four countries of interest (in order, China, France, the United States, and Japan) sampled from the fSIR parameter posterior distributions estimated via Bayesian inference plotted against true, reported SARS-CoV-2 disease transmission data.*** *Dotted lines represent sample averages for fSIR curves $s(t), i(t)$ across all timesteps $t \in T$, shaded regions represent non-parametric bootstrapped 99% confidence intervals for averages (total number of samples $M = 3000$), solid lines represent the training data $t \in T_{test}$, and scattered points represent data withheld during training $t \in T_{test}$.* ***[Center column] Plots of posterior distributions for fSIR parameters estimated in Bayesian inference****, with 'R0' corresponding to the deterministically determined random variable β/λ and 'sigma' corresponding to $\tilde{\sigma}^2$. We include the inferred parameter expectations, as well as boostrapped estimates of 95% confidence interval for each expectation, in each posterior subplot.* ***[Rightmost column] Plots of bootstrapped estimates for effective reproduction number $\mathcal{R}(t)$ (given linear infection-aware social distancing $(\kappa \circ i)(t) = \kappa \cdot i(t)$) are plotted for all $t \in T$.*** *Shaded regions correspond to 95% confidence intervals (computed directly as boostrapped sample quantiles) for the effective reproduction number $\hat{\mathcal{R}}(t)$, solid lines correspond to average boostrapped estimates, and horizontal dotted lines represent the mean, effective reproduction number averaged over $t \in T$.*

4.2. Comparative Analysis

We begin our comparative analysis of epidemiological parameters reflecting SARS-CoV-2 reported case growth as well as governmental response to known cases in China, France, the United States, and Japan by ordering these countries with respect to bootstrapped average estimates of the three critical parameters determining dynamics in the f*SIR* model under linear, infection-aware distancing: λ, $\mathcal{R}_0$, and κ. For each parameter, countries are ordered from *largest to smallest* estimated magnitude.

Infective Removal Rate Parameter $\hat{\lambda}$

1. China ($\hat{\lambda} \approx 0.110$)
2. Japan ($\hat{\lambda} \approx 6.01 \cdot 10^{-2}$)
3. US ($\hat{\lambda} \approx 6.00 \cdot 10^{-2}$)
4. France ($\hat{\lambda} \approx 5.92 \cdot 10^{-2}$)

Basic Reproduction Number $\hat{\mathcal{R}}_0$

1. US ($\hat{\mathcal{R}}_0 \approx 5.31$)
2. France ($\hat{\mathcal{R}}_0 \approx 4.67$)
3. Japan ($\hat{\mathcal{R}}_0 \approx 2.44$)
4. China ($\hat{\mathcal{R}}_0 \approx 1.35$)

Average Distancing Constant $\hat{\kappa}$

1. China ($\hat{\kappa} \approx 8670$)
2. Japan ($\hat{\kappa} \approx 7870$)
3. France ($\hat{\kappa} \approx 2520$)
4. US ($\hat{\kappa} \approx 1760$)

Recall from section (2.1) that in the f*SIR* model, larger values of $\lambda, \mathcal{R}_0$, and κ correspond respectively to the (1) shorter, average recovery times, (2) more contacts between infected and susceptible individuals at $t = 0$, and (3) more reactive infection-based social distancing policy.

Thus, based our model, we determine that of all the countries surveyed, **China's infection-based social distancing response has been the most reactive and effective over the course of the early phases of the SARS-CoV-2 pandemic**, with the shortest recovery times, the fewest estimated contacts between infected and susceptible individuals at $t = 0$, and the most reactive social-distancing policy. Moreover, Japan's response, though slightly less strict than that of China, was similarly effective at reducing the transmission of SARS-CoV-2, with fairly few contacts between infected and susceptible individuals and impacting infection-based social distancing as the pandemic was developing. In contrast, infection-based distancing policy has been far less strict in France and the United States, with significantly more contacts between those infected and susceptible estimated to have occurred even when reported SARS-CoV-2 case counts reached non-negligible levels.

To conclude, we discuss the effective reproduction number $\mathcal{R}(t)$ plots included in Fig.1, computed from posterior samples for κ and $\mathcal{R}_0$ as well as observed infectives over time $i_{obs}(t)$ for each country via Eq (6). In all surveyed countries, the average infected individual is estimated by our model to infect at most 1.5 people over the course of the last twenty days, demonstrating a broad, relative success at reducing SARS-CoV-2 transmission up to present day. Furthermore, in France and in the United States, the effective reproduction number estimated by f*SIR* has decreased more or less monotonically since cases first exceeded 1000, despite high estimates of basic reproduction magnitudes $|\hat{\mathcal{R}}_0|$. However, note that **in both China and Japan, the estimated, effective reproduction number has not monotonically decreased**, with about 80 days of (albeit very slow) increase predicted in China and 60 days of slow increase predicted in Japan; indeed, of the four pandemics examined in our analysis, China and Japan are the only two countries in which business and schools have began to slowly and partially re-open over the course of the last few weeks, with Wuhan having reopened after lockdown approximately 30 days ago and some cities throughout Japan having re-opened schools and business about 10 days ago. Though in China, most of the increase in the effective reproduction number can likely be attributed to Chinese citizens infected with SARS-CoV-2 returning to their home country from other countries with epidemics (e.g. Russia, the United States, etc.), *in Japan, the witnessed slow increase in SARS-CoV-2 effective reproduction is most likely due to partial re-opening and loosening in social distancing restrictions in response to reduced numbers of reported SARS-CoV-2 cases.* Thus, our analysis leads us to a critical thesis: **the recent decisions of governments, such as those of states throughout the United States, to *dramatically* limit social distancing policy and re-open economies will very likely prolong the first SARS-CoV-2 pandemic wave in governed regions**.

5. Conclusion

In this paper, we have developed and described an approach which combines Bayesian inference and a simple, infection-aware compartmental model of disease transmission for deducing general epidemiological characteristics of countries' responses to the SARS-CoV-2 pandemic directly from data, both with respect to regulation of disease transmission in the early phases of the pandemic and social-distancing policy reactivity to reported case numbers as the first wave of the pandemic has progressed.

Though the f*SIR* model discussed in our analysis is not well-suited for forecasting of pandemic outcomes due to its simplicity and the aforementioned medley of noise sources effecting SARS-CoV-2 data (as Franco notes in their initial paper), it provides a tractable baseline for inference-based determination of sophisticated response measures (e.g. effective reproduction number over time $\mathcal{R}(t)$), out performing its yet more simple predecessor *SIR* by capturing the reality of infection-based social distancing policy and the resultant negative feedback in reported infections [6]. Through Bayesian estimation of f*SIR* epidemiological parameters, we were able to quantitatively interpret and compare pandemic response characteristics across countries directly from disease transmission data. Lastly, we concluded with a brief discussion about the impact of infection-based social distancing policy in reducing transmission of SARS-CoV-2; as emphasized by the f*SIR* model, the reproduction number associated with some disease is a *dynamic* quantity, and thus, lifting social distancing restrictions abruptly while cases have not yet been entirely contained will indubitably lead to a rise in new infections at rates as high as the estimated $t = 0$ basic reproduction number $\mathcal{R}_0$.

One of the most pressing challenges in modern epidemiology is devising models that not only characterize important and complex aspects of true disease transmission dynamics in the 21st century, but also are easily interpretable, lucidly communicating to individuals without epidemiological training – whether typical citizens, students, parents, or politicians – the short-term and long-term influences of behaviors and policies such as social distancing. We believe that the f*SIR* infection-aware model devised by Franco is a step towards accomplishing this complex objective, successfully balancing accuracy against simplicity to stress the fluidity of pandemic disease transmission under infection-based social distancing while keeping data-based model fitting tractable.

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Appendix A. Hamiltonian Monte Carlo (HMC).

Hamiltonian Monte Carlo (HMC) is a Markov Chain Monte Carlo (MCMC) method for Bayesian inference that employs derivatives of the sampled density function to generate efficient transitions across an unknown posterior [3, 11]. Here, we include a basic explanation of the HMC algorithm.

Let θ correspond to our parameters of interest. In order to sample drawing from a density $p(\theta)$, the HMC algorithm introduces the auxiliary momentum density variables ρ. In the PyMC3 probabilistic programming framework, this auxiliary momentum density ρ is instantiated independently from the parameters of interest θ via a transformation of the parameter space that makes sampling more efficient:

$$\rho \sim \mathit{Multinormal}(0, M),$$

where M is computed under the Euclidean metric [13]. HMC then samples by drawing from the joint density

$$p(\rho, \theta) = p(\rho|\theta)p(\theta)$$

which defines a Hamiltonian

$$\begin{aligned} H(\rho, \theta) &= -\log p(\rho, \theta) \\ &= -\log p(\rho|\theta) - \log p(\theta) = T(\rho|\theta) + V(\theta), \end{aligned}$$

where the term $T(\rho|\theta)$ is referred to as the "kinetic energy" and $V(\theta)$ as the "potential energy."

HMC transitions from one set of parameter values θ_t to a new state θ_{t+1} in three successive steps. First, a new value for the momentum ρ_t is drawn (independently of θ_t, as aforementioned)

$$\rho_t \sim \mathit{Multinormal}(0, M).$$

Then, the joint system (ρ_t, θ_t) is evolved via Hamiltonian dynamics

$$\begin{aligned} \frac{d\theta}{dt} &= \frac{\partial H}{\partial \rho} = \frac{\partial T}{\partial \rho} \\ \frac{d\rho}{dt} &= -\frac{\partial H}{\partial \theta} = -\frac{\partial T}{\partial \theta} - \frac{\partial V}{\partial \theta} = -\frac{\partial V}{\partial \theta}, \end{aligned}$$

since the momentum density is independent of the target density, $\frac{\partial T}{\partial \theta} = 0$. To estimate the values of this differential equation, PyMC3 relies upon the leapfrog numerical integration algorithm, parameterized by the step-size ε:

$$\begin{aligned} \rho_t^l &\sim \mathit{Multinormal}(0, M) \\ \rho_t^l - \frac{\varepsilon}{2}\frac{\partial V}{\partial \theta} &\to \rho_t^l \\ \theta_t^l + \varepsilon M^{-1}\rho_t^l &\to \theta_t^l \\ \rho_t^l - \frac{\varepsilon}{2}\frac{\partial V}{\partial \theta} &\to \rho_t^{l+1}, \end{aligned}$$

running the above over $l \in L$ steps for $L\varepsilon$ total simulation time, eventually obtaining the values (ρ_t^*, θ_t^*). Finally, as the third step before transition, HMC accounts for possible numerical errors during integration by applying a Metropolis acceptance step, accepting the proposal values (ρ_t^*, θ_t^*) as the new transition step from (ρ_t, θ_t) with probability

$$\min\{1, \exp(H(\rho_t, \theta_t) - H(\rho_t^*, \theta_t^*))\}.$$

In the case the proposal values (ρ_t^*, θ_t^*) are not accepted, the previous parameter values (ρ_t, θ_t) are used to initialize the next iteration instead. The first N steps of sampling are referred to as the "burn-in" period, the period over which our HMC algorithm reaches a fairly stable stationary distribution; during analysis, we discard this initial sampling period, since points explored before stability is reached may correspond to relatively low probability states.

In this paper, we use the no-U-turn sampler (NUTS) to dynamically adapt the number of steps L taking by the integrator on the fly [7].